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Do important drugs reach the market sooner?

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and

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Since the Food and Drug Administration (FDA) Amendments of 1962, the average time from a drug's first worldwide patent application to its approval by the FDA has risen from 3.5 to 13.5 years. FDA policies and manufacturers' incentives suggest that more important drugs may have reached the market sooner. To test this, we develop measures of "time to approval" and "importance," and determine how the latter affects the former. Our results indicate that more important drugs are developed and approved more rapidly than less important drugs. These results imply that the costs of approval lags have probably been overstated and challenge estimates of the returns to research and development in the pharmaceutical industry.

1. Introduction

■ There is renewed concern about the stringency of Food and Drug Administration (FDA) regulation of the pharmaceutical industry, sparked in part by a desire for new drugs with which to treat AIDS. Current FDA rules can be traced to the FDA Amendments of 1962, which required the FDA to develop standards for establishing the safety and efficacy of new drugs.[1] Hansen (1979) and DiMasi et al. (1991) have documented the lengthy lags and enormous costs associated with compliance with the 1962 Amendments. Hansen estimates that the average time from initial human subject testing until final FDA approval of drugs approved during the late 1960s and 1970s was about 6 years. Dimasi et al. estimate that, by the 1980s, this process took an average of 9 years.[2] These studies and others like them have caused many researchers and policy makers to take a critical view of the 1962 Amendments.

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[1] Note that the term "drug" is used here to refer specifically to ethical drugs.

[2] In a more qualitative study, Gieringer (1985) estimates that a 1-year delay in the introduction of new drugs results in between 32,000 and 76,000 additional deaths per decade.

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Many studies find that the FDA Amendments effectively raise the cost of developing new drugs and reduce R&D productivity (Baily, 1972; Jensen, 1987; Peltzman, 1973; Wardell and Lasagna, 1975; Wardell et al., 1978; Wiggins, 1981). Most of these studies focus on drugs that represent new molecular entitities (NMEs), as opposed to novel compounds of existing drugs, or other less significant advances. As in these studies, when we discuss drugs, we specifically mean NMEs.

Previous studies measure innovative output as the total number of NMEs approved by the FDA. Although this is certainly a significant measure of innovative output, all NMEs are not of equal importance. In particular, some NMEs constitute fundamental innovations of scientific and therapeutic value, whereas others are only minor advances over existing treatments. In assessing the consequences of the FDA Amendments, this distinction is potentially of great importance. If regulation has prevented or delayed the introduction of only less important drugs, then regulation has been far less costly than was previously thought. On the other hand, if drugs of fundamental importance have been kept from reaching patients, then the costs may have been even greater than was indicated by previous studies.

In fact, both the drug industry and the FDA try to accelerate the approval of more important drugs. Our personal conversations with individuals at several drug houses suggest that drug makers have attempted to identify and accelerate development of "important" drugs for the past 30 years or more. The FDA is known to have intensified its efforts to accelerate the approval of more important drugs beginning in the mid-1970s, following pressure to do so from both Congress and the pharmaceutical industry (Kaitin et al., 1991).

This article investigates whether drugs of different importance reach the market at different rates. A related study by the Office of Technology Assessment (OTA) (1989) found that commercially important drugs have longer effective patent lives, indicating that they required less time from U.S. patent to FDA approval. The OTA study focuses on only one measure of importance—commercial importance—and examines a relatively short period of time. Our study examines multiple measures of both scientific and market importance over a long period of time. This enables us to develop more precise hypotheses regarding why important drugs reach the market sooner and to assess the relative responsibility of drug makers and the FDA in accelerating approval.

We correlate scientific and market measures of drug importance with measures of development and approval time. Our results indicate that, beginning in the 1950s, more important drugs—especially drugs that proved to be successful in the marketplace—have been developed and approved more rapidly than less important drugs. Despite this, the overall trend of increasing average development and approval times implies that even drugs two standard deviations above the mean level of importance are taking longer and longer to reach the market.

In addition to documenting the accelerated development and approval process for important drugs, we find several strands of evidence that suggest that the acceleration is due more to the actions of firms than to the FDA initiatives of the mid-1970s. However, we find no evidence that individual firm experience in dealing with the approval process is associated with accelerated review times. We conclude by challenging two critical assumptions used in the literature assessing the riskiness of pharmaceutical R&D, and test their robustness.

2. How a new molecular entity reaches the market

■ The discovery process for an NME usually begins with the identification of promising chemical and/or biological properties of a newly synthesized or previously known substance. If the research scientist is directly employed by a pharmaceutical firm, the NME may immediately enter the development phase. If the scientist is employed by a university,

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pharmaceutical firms may become aware of the discovery through a variety of channels, including research publications and presentations at scientific meetings.

The development phase of an NME begins when the pharmaceutical firm identifies a therapeutic use. At that time, the pharmaceutical firm usually applies for a patent in its home country. Under international patent agreements, the firm then has 1 year to initiate patent applications in other countries. If the firm wishes to proceed with the development of the NME, it files for approval with the regulatory agencies of each country to test and then to market the NME.

In the United States, the FDA has responsibility for approving NMEs. Under the Food, Drug and Cosmetic Act of 1938, all firms seeking to market an NME are required to file a New Drug Application (NDA), which includes proof of safety. Before 1962, the mean length of time from the filing of an NDA to its approval by the FDA was about 6 months. In 1962, following revelations that several unsafe drugs (including Thalidomide) had been approved by the FDA or were under review, Congress enacted amendments to strengthen the Food, Drug and Cosmetic Act.

The 1962 Amendments require the FDA to establish both the efficacy and safety of NMEs before granting approval to market. The drug maker's first step toward obtaining FDA approval is the filing of an Investigational New Drug Application (IND), which requests approval for a particular protocol for testing in humans. In evaluating the IND, the FDA draws on animal studies and any international clinical experience with the drug. Human testing is divided into three phases. Phase I involves evaluation of the dose-response relationship in a small number of healthy volunteers. Phase II evaluates the drug for safety and efficacy in a small number of patients who have the relevant disease. Phase III consists of broader testing with a larger number of patients, often in multiple centers and usually as part of a double-blind study. As discussed by Hansen (1979) and DiMasi et al. (1991), the average duration of trials for INDs was about 3 years for drugs that began IND testing in the period 1963–1975 and more than 6 years for drugs that began testing in the period 1970–1982. After completing IND trials, the drug goes through the traditional NDA review process, which now takes approximately 3 years.

3. Accelerating the approval of important drugs

■ In the mid-1970s, the FDA altered the approval process in order to accelerate the review of what it considered to be important drugs. To do this, it developed a system for rating INDs according to the novelty of their chemical structure (I–VI), and it also began to actively use a scheme developed in the 1950s to assess the therapeutic novelty of new drugs. Drugs are rated A for most novel to C for least novel. The assessment of structural and therapeutic novelty is made on the basis of biochemical and animal studies as well as any available foreign experience with human subjects. Drugs that rate highly along both these dimensions are given priority in the NDA review process. Applicants are given special opportunities to confer with the FDA concerning the adequacy of their protocol for testing and the implications of early clinical results. The best example of this policy is the end-of-phase II conferences, which are reserved only for INDs that the FDA classified IA or IB (Kaitin et al., 1991; Finkel, 1980).

In the only article that we know of that directly tests the effectiveness of these methods to accelerate review, Kaitin et al. (1991) found that the FDA did successfully accelerate the NDA review of NMEs, which it had given high priority (i.e., rated A), based on therapeutic novelty. We present evidence below that these high priority drugs actually took longer to reach the NDA review phase. This may be because novel drugs are more difficult to evaluate in clinical trials, or that companies spend time and resources seeking to obtain a high priority rating.

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In addition to the FDA's efforts to expedite the review of potentially important drugs, pharmaceutical firms have a number of mechanisms to accelerate testing and approval. For example, in order to meet an FDA requirement that testing be done on a given number of subjects, a firm may either work with investigators at a small number of institutions over a long period of time, or may incur the expense of working with investigators at a larger number of institutions in order to more rapidly reach the necessary sample size. Similarly, a firm might offer greater compensation to individuals willing to participate in studies of promising new drugs. Additionally, firms can allocate their most skilled personnel to critical projects and take full advantage of opportunities to interact with the FDA during the testing and approval process. The role of firm discretion in this regard was increased in 1983. At the applicant's request, the FDA now provides limited feedback on testing protocols and results even if it does not consider the drug to have great therapeutic potential.

Both the policies of the FDA and the incentives of firms suggest that more important drugs may reach the market sooner than less important drugs. To test this, we develop measures of the time to approval and importance, and estimate hazard functions to determine how the latter affect the former. We examine the changes in this relationship over time, the relative impact of market versus scientific measures of importance, and the differences in these effects between the United States and other countries, and we are able to infer whether the drug houses or the FDA played a greater role in accelerating development and review.

4. Measuring the length of time for an NME to reach the market

■ Most researchers concerned with the effects of FDA regulation on the speed of new drug development and approval have examined the length of time either from initial IND testing to NDA approval (Wiggins, 1981; Hansen, 1979; DiMasi et al., 1991) or from NDA filing to NDA approval (Kaitin et al., 1991). DiMasi et al. correctly point out that the new drug development process begins well before IND testing; roughly half of all pharmaceutical company R&D expenditures are made before the IND phase. This implies that a more appropriate measure of the full time for development would be from the initial discovery of a drug's pharmacological action until NDA approval.

This measure is preferable to IND or NDA approval times for two additional reasons. First, changes in regulation could change incentives regarding when to file IND and NDA applications, making these filing dates deceptive measures of delay due to regulation. For example, more stringent regulation might cause a firm to wait until it had more information about an NME before proceeding with a costly approval process. Thus, IND or NDA approval times might fall even though time to approval increased. The second advantage of using the date of identification, instead of the IND date, is that INDs did not exist until the mid-1960s. Therefore, any analysis that uses IND dates must be confined to later years.

We use the date of the first worldwide patent application to approximate the discovery date. Patent applications usually require identification of the proposed medical use, and it is in the interest of the company to file an application without delay in order to protect patent rights.[3] Thus, patent applications are an early indication of discovery.

One problem with patent application data is that, for many NMEs, the patents are use patents, which may be difficult to identify. A second problem is that patent application dates are not easily available for all NMEs, so that some application dates must be estimated from patent approval dates. We were able to get both patent application and patent approval dates from either a computerized patent database or U.S. patent registers for 564

[3] It seems plausible that the long times to approval might provide an incentive for firms to delay patent application somewhat in order to postpone the date of patent expiration. We raised this issue with several individuals within the pharmaceutical industry and found no one who was aware of such behavior.

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out of 760 NMEs approved during the period studied.[4] For this sample, the mean lag from patent application to patent approval was 2 years, with a standard error of 3 years. For an additional 20 drugs, we obtained the first worldwide patent approval year from the *Merck Index*. For these drugs, we estimated the application date by subtracting 2 years.

Drugs for which we could not obtain patent dates included common chemicals such as albumin, beta-carotine, and safflower oil (for which the drug houses had developed novel therapeutic uses); metals and isotopes such as Gold, AU-198, and Thallous Chloride, TL-201; and compounds. Common chemicals, metals, and isotopes represent unique classes of products that generally did not receive patents for the FDA-approved use, and so can be excluded without concern about sample selection issues. Some compounds did receive patents, so there is some concern about the specific types of compounds that did not. Excluded compounds tended to score a little lower on our measures of importance than did included compounds (compounds in general scored well below average.) We lack adequate data to test for any associated sample selection bias.

The end date can be identified as either the date of NDA approval or the first date that the drug was available for sale. Because new drugs generally reach the market quite soon after FDA approval, we use the date of approval as it is far more readily available. We obtained the NDA approval date from the FDA for all NMEs approved since 1950.

Our key dependent variable, *TIME*, is the time in years from first worldwide patent application to NDA approval.

Although the changing demands of the FDA may have caused firms to alter IND and NDA dates within the process of drug development, these dates remain interesting milestones in the development process. Analysis of these milestones may provide some insight into the causes of the changing delays, and we examine NDA application dates. We could not locate IND dates for public use, so these were not considered further.

5. Measuring the importance of NMEs

■ The importance of a drug is an elusive concept. Some drugs represent radical new breakthroughs in treatment. One example is Propranolol, the first beta-blocker. Some drugs represent advances in pharmacology, which may be of long-term research value but have limited applicability in treatment. Alpha-blockers are a good example of this. Although they proved useful in developing our understanding of the sympathetic nervous system, they were found to be of little use as a treatment for hypertension because of their unacceptable side effects. Some drugs, such as Ranitidine (Zantac), represent marginal improvements in efficacy yet are successful in the marketplace, capture large market shares, and are introduced worldwide.

Drug makers and the FDA try to assess a drug's potential importance while it is still under development, and the FDA evaluates the NMEs for therapeutic novelty. We have this measure for all drugs approved since 1978. Most drug companies attempt to forecast the sales of drugs, taking into account the novelty of the drug, the incidence of the disease that it targets, and the competition.[5] Unfortunately, we lack any measures of the firm's expectation of drug importance. We must rely on *ex post* measures (e.g., sales) and assume that these are correlated with the forecasts. To the extent that *ex post* measures of importance are only noisy measures of *ex ante* forecasts, and to the extent that companies and the FDA accelerate approvals based on *ex ante* forecasts, our estimates of differential times to approval of drugs found to be important *ex post* will understate the differential times to approval of drugs forecast to be important *ex ante*.

In recognition of the multiple meanings of importance, we have developed multiple measures. We use them individually and collectively in our analyses. Our measures are

[4] The computerized database was obtained from Robert Murawski, a patent specialist at Merck & Co.

[5] Conversation with Catherine Probst, Director of the Center for Biotechnology, Northwestern University.

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(1) citations in medical textbooks, (2) citations in medical journals, (3) citations in subsequent patent applications, (4) the extent of worldwide introduction, and (5) U.S. sales. In one analysis, we also include the FDA ranking of therapeutic novelty.

We describe below each of these variables and discuss their advantages and disadvantages as measures of importance. The measures pertain to all formulations of each NME, so that we capture their full impact.[6]

□ **Citations in medical textbooks.** One indicator of a drug's clinical importance is the amount of space that medical textbooks devote to discussing it. To construct a readily available measure of this, we examined the indexes to the 1950, 1958, 1966, 1974, 1980, and 1987 editions of *Harrison's Principles of Internal Medicine*. We tabulated the number of page references for each NME, which we call *TEXT*. We chose *Harrison's Principles* because it is a widely used text for internal medicine, the specialty that encounters the widest array of illnesses. We believe that the total number of page references is a reasonably objective measure of the clinical importance of the drug. Most references refer to specific uses of the drug, and several drugs that represent therapeutic breakthroughs (e.g., the first cephalosporins) often receive extensive discussion. Thus, a drug with three references in a given year (a score well above the mean and the median) would generally have three well-known therapeutic uses, or it may have fewer uses but be of note for its novelty.

Although some references may not relate directly to the value of the drug in treating a specific disease (e.g., some references may refer to side effects), identifying such inapplicable references would be quite cumbersome and would introduce unwanted subjectivity. Additionally, references to side effects might even be considered a measure of importance because the more widely a drug is used, the more important it becomes to inform readers of potential side effects. Ten NMEs, which play important roles in normal human physiology (i.e., insulin), were excluded because the references in texts and elsewhere were made mainly to their natural roles. This left us with a sample of 574 drugs.

□ **Citations in medical journals.** It seems reasonable that more important drugs will receive greater attention by medical researchers. Studies of efficacy and side effects are more valuable when a drug has greater potential benefits to patients. The number of articles may also reflect funding priorities by agencies such as the National Institutes of Health. The subjects of articles appearing in medical journals are indexed in *Index Medicus*. Listings in *Index Medicus* since 1962 have been computerized and may be searched using a data retrieval system called MEDLINE. We tabulated the total number of articles indexed between 1962 and 1990 for each NME, and we call this variable *MEDLINE*. There are typically hundreds of articles on each drug. Thus, we did not take on the task of reading each article and determining a measure of importance per article.[7] For NMEs that are preparations of a compound (e.g., acebutolol HCL, acetophenazine maleate), we searched under the associated *Merck Index* heading (e.g., acebutolol, acetophenazine).

□ **Citations in subsequent patent applications.** Researchers will often refer to one NME when describing the properties of another drug. The drugs may be chemically similar, or may treat the same illnesses. Patent applications in particular are likely to be thorough when identifying related drugs. Patent applications often compare the efficacy and toxicity of one drug with other drugs currently in use. If a drug is frequently cited in

[6] We also examined (1) whether a drug represented a new biological or chemical pathway as indicated by being the first drug in a new therapeutic category and (2) whether a drug appeared on the World Health Organization's *List of Essential Drugs*. An insufficient number of drugs met these criteria in later years for these measures to be useful. Including these measures in earlier years did not affect our results for those years.

[7] Counting page numbers, as we did for textbook citations, would have been prohibitively costly.

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patent applications for other drugs, it stands to reason that it is a significant standard for comparison. The texts of all U.S. patent applications for patents approved since 1975 may be searched through a data retrieval system known as LEXIS. We tabulated the number of patent applications between 1975 and 1990 that cite each NME, and we call this variable *PATCITE*.

□ **Worldwide introductions.** Grabowski (1990) argues that more important drugs will be introduced into more countries. He defined important drugs as those introduced into 6 of 11 major pharmaceutical markets: Australia, Canada, France, Germany, Italy, Japan, Norway, Sweden, Switzerland, the United Kingdom, and the United States.

Grabowski's data are for a limited time period (1970–1983), so we use an alternative measure of worldwide introductions, which we construct from data collected by DeHaen Inc. Specifically, DeHaen Inc. reports the first marketing date in each of seven major drug producing countries for all currently available NMEs.[8] We have these dates of introduction from DeHaen as of 1980, and again as of 1988. These two volumes provide worldwide introduction dates for roughly half of the NMEs approved by the FDA. We define *WORLD* as the total number of nations in which a given NME is marketed. If the drug does not appear in a DeHaen book, we let *WORLD* = 0. (When possible we calculated a variable equivalent to *WORLD* using the same source used by Grabowski. The correlation between the two measures of *WORLD* is .7.) Note that, if *WORLD* = 0, the drug is not currently available, itself an indicator of a lack of importance.

□ **Sales.** Ideally, we would like to have periodic data on worldwide sales of each drug. Unfortunately, these data appear to be available only at a prohibitive cost. Merck was able to provide us with the U.S. sales of the top 75 selling drugs in the United States for each year from 1960 to 1990. In order to construct a measure of sales using these data, we converted these numbers into 1970 dollars by discounting by the consumer price index and then summing the sales for each drug over the entire period.

Each of our measures of importance may be time dependent. For example, very old and very recent drugs may have fewer *PATCITE*s and *MEDLINE* citations than drugs developed in the 1960s and 1970s. Old drugs may not appear in the DeHaen data or in medical textbooks due to obsolescence. We control for this time dependence in two ways. First, we try to reduce biases associated with new drugs by focusing only on drugs approved no later than 1986. Second, we regress our measures of importance on a vector of period-year dummies—each dummy identifies drugs approved within a period of 6 or 7 years—and work with the residuals.

The residuals in these regressions identify the highest scoring drugs within each time period. We cannot identify any defensible way to standardize the residuals to make them comparable across time. Thus, we cannot determine whether innovative output is increasing or decreasing over time. Nor are the residuals directly comparable across measures; we address this by using Z-scores, which are constructed by dividing the residuals for each measure of importance in each period by their standard deviations. Thus, if a drug scores +2 on the measure *PATRES* (the Z-score for the residualized value of *PATCITE*), then patent citations for that drug are two standard deviations higher than the average number of patent citations for other drugs approved in its time period. This procedure implies that our measures of importance are indications of the relative rather than absolute importance of new drugs. In addition to calculating standardized residuals for each individual importance measure, we calculated the sum of standardized residuals across importance measures. We call this variable *SUMSCORE*.

[8] The seven countries are France, West Germany, Italy, Japan, Switzerland, the United Kingdom, and the United States.

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One concern in using this approach to identify the role of drug importance in accelerating approval is that causality may flow from early approval toward higher standardized importance, rather than vice versa. For example, if drug A and drug B are patented in the same year but drug A is approved earlier, it may tend to achieve higher sales and therefore higher importance, *ceteris paribus*. This might occur for two reasons. First, there may be a first-mover advantage by which the first drug of a type on the market maintains a persistent advantage over other drugs. This might occur if physicians learned the dosing and side effects of the first drug and were then reluctant to learn the corresponding information for additional drugs. The second reason a drug approved earlier might achieve greater measured importance is that it will have had more time to accumulate citations, sales, and other measures of importance.

The possibility of a first-mover advantage cannot be excluded with the data currently available. Nevertheless, it is not clear that such an effect is important. To begin with, the effect makes sense only when drugs are very close substitutes, and this is likely to be true for only a fraction of NMEs. Moreover, because becoming familiar with one drug is not a very substantial barrier to using a new drug, any new drug that has a significant advantage over an incumbent will rapidly overtake it. Additionally, there may be a novelty effect by which a newly introduced drug may be perceived as superior simply because it is new. One substantive reason for this is that, because information about a drug already on the market will be more likely to be publicly available than information about a new drug, an entrant who has access to information on both the new drug and the existing drug may be in a better position to produce clinical evidence that emphasizes the new drug's advantages. A good example in which the second mover seemed to have an advantage is provided by the antiulcer drug Zantac, which has somewhat fewer side effects and markedly higher sales than the first-mover Tagamet.

Irrespective of a first-mover advantage, there is also the possibility that a more rapidly approved drug may achieve greater measured importance simply because of the greater amount of time it has had to accumulate sales, citations, etc.[9] To address this, we performed an additional analysis in which we included within-period linear time trends in our estimates of importance. This controls for the possibility that early introduction within a period will lead to a higher importance score. The time trends were small and generally insignificant, suggesting that the problem of increased importance scores due to earlier approval within a period is relatively unimportant. Also, we allowed our period lengths to equal 1 year. This greatly increases the noise in our measures of relative importance, because the comparison group in a given year is usually less than 20 drugs. Nevertheless, we obtained qualitatively similar results to those reported in this article.

An additional dimension of this problem is that drugs that are more rapidly approved will have longer times remaining on their patents and therefore higher expected sales.[10] This is most likely to be an issue for recent years in which the approval process may often consume a drug's entire patent life.[11] To control for this, we recalculated our sales measure to look only at sales in the first 5 or 6 years after FDA approval. This is a noisy measure of total sales for two reasons. First, some drugs, such as the first cephalosporins, do not sell well until after they have been on the market for a few years, whereas others, such as Zantac, do well right away. Second, our sales data only cover the top 75 sellers, so are likely to miss many drugs that are on the rise in the first few years after their release.

[9] Note that a drug may not necessarily have higher standardized importance because its accelerated review may place it in an earlier time period.

[10] The same might be said for *MEDLINE* citations and other measures of importance. Unfortunately, it was prohibitively costly to construct time-dependent measures of importance along these other dimensions.

[11] The Patent Term Restoration Act of 1984 guarantees a minimum 7 year postapproval patent life to all recently approved drugs.

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Even so, using this measure as part of our aggregate measure of importance, we obtained results that were qualitatively similar to those reported in this article.

A final issue relates to the grouping of drugs by the date of approval, as opposed to the date of patent application. Variations over time in the difficulty of publication or stringency of FDA regulation will affect a drug's measures of importance and approval time differently depending both on the date of patent and the date of final approval to market. It therefore seems to us that neither of these measures is ideal for grouping drugs into periods. Nevertheless, we also reestimated the regressions by basing our periods on patent application dates rather than NDA approval dates. Thus, in unreported regressions, we estimated *TIME* for all drugs discovered within a given time period.[12] Reclassifying drugs in this manner was not found to significantly alter our results concerning the effect of importance.

6. Empirical approach

■ Kiefer (1988) notes that, when the model generating the dependent variable is represented as a sequence of conditional probabilities, the model is best analyzed by estimating hazard functions. The process of bringing a new drug to market appears to fit the model of a sequence of conditional events, in which entry into each successive stage of the R&D process is conditional on successful completion of the previous stage. Thus, hazard analysis is appropriate for estimating and interpreting the data.[13]

The hazard rate for a given duration t (i.e., $t = TIME$) equals the rate at which drugs will be approved at duration t, given that they have not been approved before t. Let $H(t)$ denote the hazard rate and T the time of approval. Then,

$$H(t) = \lim_{h \to 0} \text{Prob}[t \leq T \leq t + h \mid T \geq t]. \tag{1}$$

We will estimate a Weibull parameterization of the hazard rate. Specifically, we will assume that

$$H(t) = \lambda p(\lambda t)^{p-1}, \tag{2}$$

where $\lambda = e^{\beta_0 + \underline{\beta}\mathbf{X}}$ and $p = 1/\sigma$, $\underline{\beta}$ and σ are parameters to be estimated, and $\mathbf{X}$ is a vector of covariates (e.g., measures of importance). This parameterization has as a special case ($\sigma = 1$) the situation in which the hazard rate is independent of duration. The parameters are estimated by maximum likelihood. Estimated standard errors may be biased if there are omitted variables (see Kiefer, 1988). We checked for this using a technique described by Greene (1990). We were able to reject the presence of this bias in all time periods.

The parameter estimates may be interpreted as follows. The value of σ determines if the hazard rate is declining or increasing with duration. If $\sigma < 1$, for example, the hazard is increasing in duration, so that the conditional probability of drug approval increases over time. The values of σ and λ jointly determine the slope of the survivor function, which relates the probability of surviving (i.e., not being approved) to duration. From

[12] There may be truncation bias because some patented drugs that will eventually be approved have not yet been approved. To assess the possible bias of the coefficient on *SUMSCORE* due to this, we compared this coefficient in two restricted samples: (1) all drugs patented before 1965 and (2) all drugs patented before 1965 and approved by 1980. Although both samples are truncated, the truncation is obviously greater for the second sample. If truncation bias is important, the results from the first regression should differ substantially from those in the second. In fact, the coefficient on *SUMSCORE* is nearly the same in both cases, suggesting that truncation bias is minimal.

[13] Hazard estimates are especially appropriate in periods 1 and 2, when the dependent variable has a relatively high proportion of values near zero. OLS estimates are similar to the hazard estimates, particularly for periods 3–6.

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this, we can easily calculate the distribution of durations for drugs with any given values of the covariates **X**. Specifically, the $P/100$th percentile approval time is given by

$$[(-\log P)^{\sigma}]/\exp[\beta_0 + \beta_x \mathbf{X}]. \tag{3}$$

To illustrate the calculation of estimated approval times, suppose that the covariate is *SUMSCORE*. Then the median (50th percentile) approval time of drugs with average importance (*SUMSCORE* = 0) is $(-\log(.5))^{\sigma}/\exp(\beta_0)$. An important implication of this formula is that, for small values of $\beta_{\mathbf{X}}$, if the value of the covariate **X** increases by one unit, then the median approval time increases by a factor of $\beta_{\mathbf{X}}$. For example, if $\beta_{SUMSCORE} = -.05$, then a drug that scores +1 on the variable *SUMSCORE* will have a median approval time that is 5% less than a drug that scores 0.

7. Results

■ Table 1 shows the time dependence of unadjusted measures of importance. For example, all of the raw importance scores decline in period 6. The mean of *SUMSCORE* is zero in each period as a result of our standardization procedure. The standard deviation of *SUMSCORE* differs across periods because the components that make up *SUMSCORE* are not equally correlated across periods. *TIME*, which equals the mean time from first international patent to FDA approval, declines slightly before the 1962 Amendments, followed by the well-recognized pattern of increases since that time. The average *TIME* increases to almost 6 years by the mid-1960s and continues to rise until reaching more than 14 years by the mid-1980s.

In Table 2, all five standardized importance measures are significantly positively correlated at $p < .001$. Note that the correlations between the three scientific measures (*PATRES*, *MEDRES*, and *TEXTRES*) are all greater than .6, whereas the market-based measures

TABLE 1 Summary Statistics for Entire Sample, by Period

Variable Name	Full Sample 1950–1986	Period 1 1950–1955	Period 2 1956–1961	Period 3 1962–1967	Period 4 1968–1974	Period 5 1975–1980	Period 6 1981–1986
PACITE	106	83	95	220	126	86	65
	(182)	(198)	(131)	(341)	(166)	(95)	(81)
MEDLINE	1316	873	1027	2500	1760	1307	990
	(2642)	(2295)	(2253)	(4488)	(2696)	(1832)	(1402)
TEXT	3.51	4.18	2.85	6.65	3.93	3.06	1.84
	(6.95)	(10.11)	(6.63)	(10.88)	(5.81)	(3.87)	(2.65)
SALES	1604	5.35	1481	2878	1220	2278	1571
	(5234)	(2260)	(4346)	(9076)	(4004)	(6269)	(4336)
WORLD	2.74	.86	1.73	3.31	3.67	4.37	3.37
	(2.28)	(1.11)	(1.76)	(2.17)	(2.49)	(1.91)	(2.19)
TIME	7.99	3.91	3.41	5.92	9.63	12.3	14.04
	(6.54)	(4.25)	(4.10)	(4.78)	(4.79)	(5.53)	(6.06)
SUMSCORE	0	0	0	0	0	0	0
	(3.61)	(3.76)	(3.09)	(6.37)	(3.19)	(2.74)	(2.36)
SUMSCI	0	0	0	0	0	0	0
	(2.62)	(3.28)	(2.27)	(4.67)	(2.19)	(1.44)	(1.18)
SUMMKT	0	0	0	0	0	0	0
	(1.56)	(.78)	(1.31)	(2.30)	(1.61)	(1.73)	(1.62)
N	574	91	140	72	75	82	114

Standard deviations are in parentheses.

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TABLE 2 Correlation Matrix of Standardized Importance Measures

	PATRES	*MEDRES*	*TEXTRES*	*SALESRES*	*WORLDRES*
PATRES	1.00	.62	.64	.47	.26
MEDRES	.62	1.00	.69	.37	.21
TEXTRES	.64	.69	1.00	.34	.25
SALESRES	.47	.37	.34	1.00	.23
WORLDRES	.26	.21	.25	.23	1.00

Note: All correlations are significant at $p < .001$.

(*SALESRES* and *WORLDRES*) are less strongly correlated both with the scientific measures and with each other. Examining a sample of 192 drugs introduced in the 1950s–1970s, Parker (1984) also finds a positive correlation between scientific importance (based on evaluations by pharmacologists) and sales and introductions into 17 nations.

Although it would probably be impossible to develop a consensus from a group of health professionals on a list of the most important drugs, Table 3 shows a collection of drugs that individuals knowledgeable about pharmaceuticals would clearly consider extremely important. It is interesting to note that these drugs represent a variety of therapeutic categories both within and across periods.

In Table 4 we estimate a separate hazard equation for each time period. This permits flexibility, for example, by allowing σ to vary across periods. The key result in Table 4 is that $\beta_{SUMSCORE}$ is negative and significant in periods 2–6. This implies that important drugs had speedier development and approval as long ago as the mid-1950s. We should point out that $\beta_{SUMSCORE}$ declines in magnitude and is no longer significant in period 3 when we include covariates indicating the use of the drug (see Table 8). However, these covariates tend to be insignificant, suggesting that the results of the more parsimonious model are preferable.

Other findings are that β_0 has increased substantially over time and σ has decreased over time. The increase in β_0 is consistent with the observed increase in average *TIME*. The results that $\sigma < 1$ in all periods and that σ has decreased over time imply that (1) approval rates increase with duration in all periods and (2) the increase is more dramatic in later periods. This latter result reflects the substantial fixed hurdles that make very rapid approvals much more difficult in later years.

In Table 5 we estimate approval times for drugs of average *SUMSCORE*, as well as drugs for which *SUMSCORE* is one standard deviation above and below average, and two standard deviations above average. The latter represent what we might consider breakthrough drugs. The median approval time for average drugs shows the characteristic large increase over time. Approval times increase for drugs of all importance levels, although the rise is not as great for the most important drugs. There is a substantial difference in approval times between the most important drugs and less important drugs. A two standard deviation positive outlier reached the market roughly 2 years sooner than a drug of average importance in periods 3 and 4 (roughly the 1960s and early 1970s) and about 3 years sooner in periods 5 and 6 (the late 1970s and early 1980s). This translates into a 25% reduction in development and approval times for the most important drugs. These results are illustrated in Figure 1. Interestingly, the OTA found that the highest selling drugs approved by the FDA between 1985–1989 had effective patent lives that were 3.3 years longer than the lowest sellers—very close to our estimates for the early 1980s. (The effective patent life is almost an exact complement to *TIME*. Differences stem from the variation between the first worldwide patent application and U.S. patent approval.)

Important drugs obtained more rapid development and approval as early as the 1950s, well before the FDA undertook efforts to accelerate the approval of important drugs in the mid-1970s. Measured in percentage terms, the rate of acceleration of important drugs

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TABLE 3 Six Most Important Drugs with Highest Values for *SUMSCORE*, by Period

	SUMSCORE	Drug	Therapeutic Category
Period 1	25.08	Chloramphenicol	Antibiotic
	12.64	Chlorpromazine (Thorazine)	Antipsychotic
	12.11	Phenylbutazine	Anti-inflammatory
	5.53	Phentolamine	Cardiovascular (α-Blocker)
	5.05	Meprobamate (Miltown)	Sedative-Hypnotic
	4.43	Hydralazine	Cardiovascular (Vasodilator)
Period 2	16.43	Cyclophosphamide	Antineoplastic
	11.93	Chlordiazepoxide (Librium)	Anxiolytic
	8.24	Dexamethasone	Corticosteroid
	7.68	Methylprednisolone	Corticosteroid
	7.52	Propoxphene (Darvon)	Opioid Analgesic
	7.34	Amitriptyline (Elavil)	Tricyclic Antidepressant
Period 3	24.36	Propranolol	Cardiovascular (β-Blocker)
	23.13	Diazepam (Valium)	Anxiolytic
	22.32	Indomethacin	Anti-inflammatory
	14.22	Gentamicin	Antibiotic (Aminoglycoside)
	8.53	Furosemide (Lasix)	Diuretic (Loop Diuretic)
	8.43	Cephalothin	Antibiotic (I-G Cephalosporin)
Period 4	9.15	Ibuprofen (Advil, Motrin)	Anti-inflammatory
	9.04	Cephalexin	Antibiotic (I-G Cephalosporin)
	7.20	Azathioprine	Immunosuppressant
	6.81	Naloxone	Opioid Antagonist
	6.12	Rifampin	Antibiotic Antiviral
	5.86	Carbenicillin	Antibiotic (Penicillin)
Period 5	12.56	Cimetidine (Tagamet)	Antisecretory (Peptic Ulcer)
	5.30	Naproxen	Anti-inflammatory
	5.13	Cisplatin	Antineoplastic
	4.91	Tobramycin	Antibiotic (Aminoglycoside)
	4.87	Cefoxitin	Antibiotic (2-G Cephalosporin)
	4.85	Metoprolol (Lopressor)	Cardiovascular (β1 Blocker)
Period 6	9.04	Ranitidine (Zantac)	Antisecretory (Peptic Ulcer)
	8.82	Nifedipine	Cardiovascular (Ca Channel Blocker)
	6.57	Captopril	Cardiovascular (ACE Inhibitor)
	6.30	Atenolol	Cardiovascular (β1 Blocker)
	5.40	Diltiazem	Cardiovascular (Ca Channel Blocker)
	4.76	Piroxicam	Anti-inflammatory

Note: Only those drugs for which *TIME* could be calculated were included.

has not changed in any systematic way since period 2. Measured in absolute terms, however, the magnitude of acceleration has increased since the mid-1970s. One explanation for these findings is that, as average review lags lengthened, firms increased their efforts to further accelerate review. Of course, the efforts of the FDA may also play a role in the accelerated review since period 4.

As discussed, the FDA has undertaken a number of actions to try to accelerate approval of certain drugs. One of the actions for which we were able to obtain relevant data is the assignment of a therapeutic novelty rating of A, B, or C to drugs in the NDA application phase. The FDA reports these rankings for all drugs approved in periods 5 and 6. We reestimated the hazards for periods 5 and 6, including *SUMSCORE* and indicator variables for the FDA rankings. As reported in Table 6, we found that the coefficients on *SUMSCORE* were only slightly lower when we included the FDA rankings and that the coefficients on the FDA rankings were insignificant and of variable sign. This suggests that marketing importance may have more to do with the acceleration of drug approval than scientific importance.

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TABLE 4 Hazard Equation Results
Dependent Variable = *TIME*; Covariate = *SUMSCORE*, by Period

	Period					
Coefficients	1	2	3	4	5	6
$\beta_{SUMSCORE}$	−.011	−.058*	−.032**	−.037**	−.060*	−.045*
	(.050)	(.024)	(.015)	(.018)	(.022)	(.017)
β_0	1.54	1.33	1.89	2.38	2.61	2.76
σ	.78	.87	.66	.45	.40	.39
No. of observations	91	140	72	75	82	114

Note: Standard errors of $\beta_{SUMSCORE}$ are in parentheses. The coefficient β_0 is significantly larger than zero in all periods at $p < .001$, and the coefficient σ is significantly smaller than one in all periods at $p < .05$.
* Significant at the .01 level.
** Significant at the .05 level.
*** Significant at the .10 level.

To further examine the relative strength of marketing and scientific importance, we broke up *SUMSCORE* into two components. We combined *PATRES*, *MEDRES*, and *TEXTRES* into a measure of scientific importance, which we call *SUMSCI*. We combined *SALESRES* and *WORLDRES* into a market oriented measure of importance, which we call *SUMMKT*. Table 7 reports the hazard function coefficients when we replace *SUMSCORE* with *SUMMKT* and *SUMSCI*. In all periods, the coefficient on *SUMMKT* is substantially larger than the coefficient on *SUMSCI*. The *SUMMKT* coefficients are significant in 4 of the 6 periods, including periods 4–6, whereas the coefficients on *SUMSCI* are never significant.

This result can be interpreted in several ways. If we believe that drug companies are more concerned with the market performance of a drug than is the FDA, and that market success *ex post* is correlated with predicted success *ex ante*, this indicates a greater role for drug companies in the acceleration of important drugs. On the other hand, if the FDA acts to maximize consumer surplus and *SUMMKT* is a better proxy for expected consumer surplus than is *SUMSCI*, the result could be consistent with a greater role for the FDA. Alternatively, it could be that drugs of greater scientific importance might be more difficult to evaluate during the FDA review process, by virtue of novel mechanisms of action with perhaps a new or unexpected range of side effects.

TABLE 5 Estimated Median Times to Approval for Various Values of *SUMSCORE*, by Period

	Period					
	1	2	3	4	5	6
2 standard deviation above mean *SUMSCORE*	3.25	1.94	3.46	7.24	8.39	11.03
1 standard deviation above mean *SUMSCORE*	3.38	2.30	4.25	8.14	9.94	12.27
Mean *SUMSCORE*	3.51	2.74	5.20	9.16	11.78	13.66
1 standard deviation above mean *SUMSCORE*	3.66	3.27	6.43	10.31	13.96	15.20

Note: All values expressed as years from first patent application to FDA approval.

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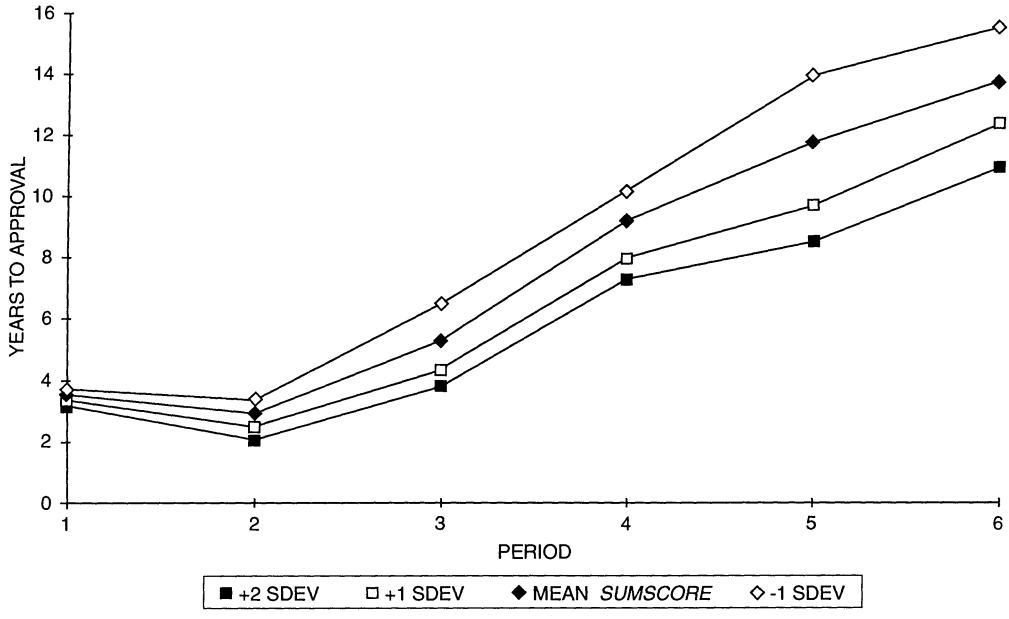
FIGURE 1

MEDIAN TIME TO APPROVAL BY *SUMSCORE* AND PERIOD



□ **Drug and firm characteristics.** There are several reasons to expect that the length of time required to bring a drug to market may vary with its proposed use. For example, a drug intended for a chronic condition generally requires lengthy testing in order to evaluate long-term efficacy and toxicity. Likewise, regulators may be more cautious in approving drugs intended for children or women of reproductive age, and it may be more difficult to find very young (or very old) volunteers for clinical trials. These examples suggest the importance of controlling for the proposed use of a drug when estimating times to approval.

Although there are many attributes of a drug's proposed use that could influence the time to approval, we focus on the age of the target population and the extent of chronic

TABLE 6 Hazard Equation Results
Dependent Variable = *TIME*; Covariates = FDA Therapeutic Novelty Ratings

	Period	
Coefficients	5	6
$\beta_{SUMSCORE}$	−.056*	−.045*
	(.02)	(.02)
$\beta_{A\text{-rating}}$	−.253	.249
	(.21)	(.17)
$\beta_{B\text{-rating}}$	−.104	−.019
	(.09)	(.07)
β_0	2.70	2.75
σ	.38	.38

Note: Standard errors are in parentheses. The coefficient β_0 is significantly larger than zero in all periods at $p < .001$, and the coefficient σ is significantly smaller than one in all periods at $p < .05$.

* Significant at the .01 level.

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TABLE 7 Hazard Equation Results
Dependent Variable = *TIME*; Covariates = *SUMMKT* and *SUMSCI*, by Period

	Period					
Coefficients	1	2	3	4	5	6
β_{SUMMKT}	−.038	−.113***	−.052	−.118*	−.106*	−.071**
	(.160)	(.066)	(.050)	(.027)	(.036)	(.032)
β_{SUMSCI}	−.007	−.041	−.025	.030	−.008	−.011
	(.054)	(.031)	(.029)	(.027)	(.033)	(.046)
β_0	1.54	1.33	1.89	2.38	2.61	2.76
σ	.78	.86	.66	.41	.38	.38

Note: Standard errors are in parentheses. The coefficient β_0 is significantly larger than zero in all periods at $p < .001$, and the coefficient σ is significantly smaller than one in all periods at $p < .05$.
* Significant at the .01 level.
** Significant at the .05 level.
*** Significant at the .10 level.

versus acute use. (There are not enough "male only" or "female only" drugs to test for related differences in *TIME*.) We expect these may be of particular importance, and they also have the advantage of being relatively easy to identify for a drug of a given therapeutic use. Specifically, we define the following variables.[14]

CHRONIC = 1 if a drug was judged to be primarily for chronic use.
ACUTE = 1 if a drug was judged to be primarily for acute use (the omitted category is drugs intended for both acute and chronic use).
YOUNG = 1 if a drug was judged to be primarily for use by the young.
OLD = 1 if a drug was judged to be primarily for use by the elderly (the omitted category is drugs intended for both the young and old).

In our sample, 233 drugs scored 1 for *CHRONIC*, 227 scored 1 for *ACUTE*, and 114 scored 1 for both. 33 scored 1 for *YOUNG*, 117 scored 1 for *OLD*, and 424 scored 1 for both. In unreported analyses, we also controlled for therapeutic category (e.g., cardiovascular). The results were not altered by inclusion of these category effects.

In addition to these measures of drug characteristics, we considered one additional covariate to reflect the size of the firm that makes the drug. Thomas (1990) suggests that the testing requirements of the 1962 FDA Amendments may have benefited large, research-driven firms relative to smaller, imitative firms. One possible reason is that large firms benefit from scale economies in performing drug testing. For example, large firms may, through experience, develop relationships with many research physicians that foster quicker and more effective drug testing. Another possible reason is that large firms may produce better selling drugs. If the costs of obtaining FDA approval are largely independent of a drug's potential sales, then testing creates an economy of scale that is more likely to benefit large firms.

To test these ideas, we constructed a measure of a firm's experience in dealing with the FDA review process. Our data restrict us to assigning each drug to the firm that filed the NDA. We measure a firm's *EXPERIENCE* for a given drug as the cumulative number of FDA approvals in the periods before the approval period of the drug. For example, if

[14] These variables were defined according to the drug's primary 5-digit therapeutic classification as identified by IMS America. For example, the IMS category consisting of beta-blockers for the treatment of cardiovascular diseases was judged to be intended primarily for chronic use by older patients. A list of these classifications is available from the authors upon request. We also tried indicator variables for specific drug categories (e.g., anti-infectives) with comparable results.

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a Lilly drug X was approved in period 4, then *EXPERIENCE* equals the number of Lilly drugs approved in periods 1–3. *EXPERIENCE* = 0 for all drugs approved in period 1.[15]

Table 8 reports the hazard function coefficients when we include our measures of drug and firm characteristics. Firm *EXPERIENCE* is not a significant predictor of approval times. The coefficient is generally in the expected direction but is small and insignificant.[16] The coefficients on drug characteristics *YOUNG* and *CHRONIC* show no consistent pattern. Some evidence that drug characteristics matter may be obtained by considering the coefficients on *OLD* and *ACUTE* in periods 4–6, the three periods when the IND testing requirements became most onerous. The three coefficients on *OLD* have the expected sign and are jointly significant at $p < .10$.[17] The same is true for the three coefficients on *ACUTE*. Based on the point estimates, we can conclude that a drug targeting *ACUTE* illnesses will reach the market 10–20% sooner than a drug targeting both acute and chronic illnesses. Similarly, a drug targeting *OLD* patients will reach the market 10–20% later than a drug targeting patients of all ages.

□ **Decomposition of the time to approval.** There are several important milestones between the first patent application and final approval by the FDA. The effects of importance on *TIME* may be felt between various milestones. Examining approval times by subperiod may reveal where the approval process slows down as well as indicate specific ways in which the approval of important drugs has been accelerated.

TABLE 8 Hazard Equation Results
Dependent Variable = *TIME*; Covriates = Drug and Firm Characteristics

	Period					
Coefficients	1	2	3	4	5	6
$\beta_{SUMSCORE}$	−.003	−.056***	−.023	−.042*	−.064*	−.066*
	(.04)	(.03)	(.02)	(.02)	(.02)	(.02)
$\beta_{EXPERIENCE}$	—	−.036	−.022	−.005	−.005	.003
		(.03)	(.02)	(.01)	(.006)	(.004)
β_{YOUNG}	.491	.335	.230	−.333	.029	—
	(1.40)	(.30)	(.39)	(.25)	(.34)	
β_{OLD}	−.347	−.083	.132	.112	.221	.140
	(.34)	(.23)	(.31)	(.21)	(.16)	(.14)
β_{ACUTE}	−.085	.076	.466	−.020	−.265***	−.134
	(.22)	(.21)	(.35)	(.19)	(.16)	(.19)
$\beta_{CHRONIC}$	−.019	−.088	.124	.133	−.249	.026
	(.22)	(.26)	(.36)	(.23)	(.19)	(.19)
β_0	1.62	1.47	1.80	2.33	2.84	2.72
σ	.76	.85	.61	.43	.38	.37

Note: Standard errors are in parentheses. The coefficient β_0 is significantly larger than zero in all periods at $p < .001$, and the coefficient σ is significantly smaller than one in all periods at $p < .05$.

* Significant at the .01 level.

** Significant at the .05 level.

*** Significant at the .10 level.

[15] Economies may be derived from overall size, rather than experience. We cannot determine overall pharmaceutical sales for each firm. However, examination of the firms with high and low values for the variable *EXPERIENCE* suggests that this variable would be a suitable proxy for overall size.

[16] It is possible that large firms economize by reducing their expenditures per year of FDA review rather than reducing the length of review. This would be unlikely in the case of a drug with large anticipated sales, but cannot be ruled out in general. We lack data on total and annual testing expenditures, so we cannot evaluate this conjecture.

[17] Based on a χ^2 test. For details, see Maddala (1977).

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TABLE 9 Mean Values of *TIME1* and *TIME2* by Period

	Period					
	1	2	3	4	5	6
TIME1	3.89	2.89	4.05	7.51	9.77	12.28
TIME2	.32	.58	2.17	2.69	2.94	3.03
No. of observations	61	128	59	59	66	88

We were able to obtain the NDA application dates for the majority of drugs in our sample. We defined *TIME1* as the time from first worldwide patent application to NDA application, and we defined *TIME2* as the time from NDA application to NDA approval. Because we could not obtain NDA application dates for many drugs, our total sample was reduced to 461 drugs. Mean values of *TIME1* and *TIME2*, by period, are reported in Table 9. As seen in the table, both phases have increased in length over the years; however, the period before NDA application is substantially longer than the period of FDA review of the NDA. We recalculated *SUMSCORE* by period for those drugs for which we knew the values of *TIME1* and *TIME2*. This implies that the mean value of *SUMSCORE* in the analyses described below is again zero.

Estimates of hazard function coefficients appear in Table 10. The results suggest that importance generally affects both phases of the drug development process. The rough comparability of the *SUMSCORE* coefficients in the table indicates that an important drug is accelerated by a comparable percentage in each phase of the review process. Because the period before NDA filing is much longer than the period of NDA review, this implies that most of the acceleration of important drugs, in absolute terms, occurs before filing the NDA. Thus, the finding of Kaitin et al. (1991) that more important drugs are accelerated through the NDA review process probably does not constitute the most important dimension of the overall acceleration of important drugs to market.

We included indicators of FDA novelty ranking in separate unreported estimates of the period 5 and 6 hazards. Interestingly, we found that A-rated drugs had shorter *TIME2*'s, consistent with Kaitin et al., but that these were offset by longer *TIME1*'s.

TABLE 10 Hazard Equation Results, by Period
Covariate = *SUMSCORE*

	Dependent Variable						
	TIME1			*TIME2*			
Period	$\beta_{SUMSCORE}$	β_0	σ	$\beta_{SUMSCORE}$	β_0	σ	Number of observations
1	−.018	1.54	.84	−0.47	−1.04	.91	61
	(.064)			(.072)			
2	−.055*	1.15	.97	−.058	−.49	.85	128
	(.020)			(.036)			
3	−.027	1.57	.81	−0.25	.88	.70	59
	(.021)			(.025)			
4	−.024	2.18	.53	−.063*	.91	.71	59
	(.024)			(.023)			
5	−.048*	2.43	.44	−.011	1.12	.84	66
	(.017)			(.046)			
6	−.036***	2.63	.44	−.047	1.21	.67	88
	(.021)			(.035)			

Note: Standard errors are in parentheses.

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8. International comparisons

■ Another way to understand the role of the FDA is by comparing U.S. approval dates with approval dates in other countries. Because drug companies are largely multinational, changes over time in their efforts to accelerate drug approvals would seem more likely to be parallel across countries.[18] In contrast, the efforts of individual countries to accelerate the approval of important drugs would likely manifest themselves in differences in the rate of approval of important drugs across countries.

Because the discovery date—first worldwide patent filing—is the same for all countries, a comparison of approval dates provides direct evidence on comparative approval lags. We compare approval dates of drugs in the United States with approval dates for the same drugs in the United Kingdom, Germany (at that time still West Germany), and France. The data on drug introductions come from the DeHaen International database, which was used to construct the *WORLD* variable. This database is restricted to currently marketed drugs; it is therefore incomplete, especially for older drugs. DeHaen reports U.S. marketing dates for only 64% of the NMEs approved by the FDA since 1950. We restrict our attention to drugs approved by the FDA in 1962 or later (i.e., periods 3–6). DeHaen reports U.S. marketing dates for 73% of these drugs. We further restrict our attention to drugs marketed in both of the countries being compared. This reduces the samples considerably.

We estimate the lag from approval in a given country to approval in the United States by differencing the appropriate marketing years. For example, *USLAGFR* is the number of years from approval in France to approval in the United States. *USLAGUK* and *USLAGWG* are defined analogously. Although approval years and marketing years do not always coincide, there is no particular reason to expect that the time from approval to marketing should vary across countries.[19] We reestimate *SUMSCORE* for the restricted samples, so that the mean value of *SUMSCORE* is always zero.

The process that generates *USLAGFR*, etc. does not lend itself to modelling as a hazard process (e.g., the values can plausibly be negative). We therefore estimate these using ordinary least squares. We include period-specific dummy variables and period-specific *SUMSCORE* effects. We include drug-type effects as covariates because these appear to be marginally significant in periods 3–6. The regression results appear in Table 11. The upward trend in the coefficients on P3, P4, P5, and P6 indicates that the lags to approval in the United States have grown progressively longer relative to other countries.

The coefficients on importance are small and generally insignificant. This implies that the accelerated approval of important drugs was a worldwide phenomenon. Assuming that regulatory change with respect to importance did not occur at the same time across countries, this further reinforces the idea that actions of drug companies are more important in the acceleration of important drugs. These findings complement those reported by Parker (1984) in his study of the diffusion of 192 new drugs across nations. He reports that the time between approval in different nations was smaller for more important drugs; here, importance is based on scientific measures. This finding is not inconsistent with our finding that approval is everywhere accelerated for important drugs.

Another interesting indicator of the importance of firm actions in the approval process is the tendency for U.S.-made drugs (*USMAKE*) to be approved 2 to 3 years earlier in the United States than they are in France, the United Kingdom or West Germany. These drugs are also approved earlier than drugs made in other countries. Similarly, French-

[18] This does not imply that companies will introduce their drugs at the same time in each country, only that efforts to accelerate drugs are likely to have similar benefits in each country.

[19] We replicate this analysis with a restricted sample of drugs for which we have overseas regulatory approval dates. (The sample is drugs first approved worldwide between 1970 and 1983.) The results are virtually identical to those reported here.

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TABLE 11 Regression Results
Dependent Variable = U.S. versus Foreign Approval Times

Dependent Variable Measure (= IM)	*USLAGFR SUMSCORE*	*USLAGUK SUMSCORE*	*USLAGWG SUMSCORE*
IM*P3	−.03	−.37*	.25***
	(.13)	(.13)	(.14)
IM*P4	.27	−.29	.36
	(.32)	(.32)	(.29)
IM*P5	−.01	−.28	−.11
	(.25)	(.27)	(.28)
IM*P6	.19	.23	−.28
	(.30)	(.31)	(.32)
ACUTE	−1.24	−.21	−1.38
	(.97)	(1.08)	(1.12)
CHRONIC	−2.13***	−.89	−1.23
	(1.09)	(1.19)	(1.26)
YOUNG	4.45**	.87	2.36
	(1.85)	(1.69)	(2.07)
OLD	2.34**	.94	.84
	(1.06)	(1.11)	(1.19)
USMAKE	−2.21*	−1.66**	−3.32*
	(.70)	(.82)	(.81)
FOREMAKE	−4.35*	−1.43	2.59
= FR, UK, WG	(1.51)	(1.24)	(1.65)
P3	−.78	−.67	.21
	(1.17)	(1.30)	(1.36)
P4	.86	1.17	2.78**
	(1.12)	(1.24)	(1.27)
P5	3.02*	1.98***	3.95*
	(1.04)	(1.18)	(1.22)
P6	2.67**	3.05**	3.46*
	(1.12)	(1.23)	(1.28)
R^2	.26	.16	.19
N	198	193	220

* Significant at the .01 level.
** Significant at the .05 level.
*** Significant at the .10 level.

made drugs are clearly at a dramatic advantage for approval within France, just as German-made drugs have a large (though marginally insignificant) advantage in Germany. The "home court advantage" may reflect firm expectations of higher sales at home, which translate into efforts to accelerate approval. Or, they may reflect a "home court bias" of regulators.

9. Implications for estimates of the returns to R&D

■ There is extensive literature that estimates the returns to R&D in the pharmaceutical industry (Ayanian, 1974; Friedman and Associates, 1973; Grabowski and Vernon, 1990; Joglekar and Patterson, 1986; Virts and Weston, 1980). Our findings challenge two critical assumptions used in these articles and allow us to test their robustness.

The first assumption, used in all of these articles, is that the cost of bringing a new drug to market is identical for all drugs. If development costs are related to approval times, then our research suggests that costs may vary systematically by drug importance. Using simulated R&D costs reported in DiMasi et al. (1991), the fully capitalized cost of bringing to market drugs that score in the 95th percentile in *SUMSCORE* may be as much as 20% lower than the cost of bringing an average drug to market. This suggests that the variance

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of the rates of return on new drugs is higher than generally reported. However, given the tremendous skewness of the distribution of drug sales, the additional variance attributable to differences in development costs may be relatively small. (See Grabowski, 1991, for a discussion of the variation in drug sales.)

The second assumption that is often used to estimate rates of return is that worldwide drug sales can be estimated from U.S. sales by assuming that the ratio of worldwide sales to U.S. sales is equal for all drugs (Grabowski and Vernon, 1990; Joglekar and Patterson, 1986; Virts and Weston, 1980). The positive correlation between *WORLDRES* and *SALESRES* suggests that drugs that sell better in the United States are marketed in more foreign countries and should therefore have a greater multiplier. This effect is likely to be relatively small, however. The correlation in our sample is only .23, suggesting that a two standard deviation outlier in sales would be introduced in less than one additional country.

10. Conclusions

■ Although the average approval time for drugs has risen by about 10 years since the FDA Amendments of 1962, our results indicate that the average approval time of drugs two standard deviations above the mean in importance has risen by only about 7 years. This implies that estimates of the FDA Amendments' cost based on the average increase in delay substantially overstate the true costs of the Amendments. For example, Gieringer's (1985) rough estimate that a 1 year increase in the delay in introducing new drugs results in 32,000 to 76,000 deaths per decade assumes that the increased delay applies equally to all drugs. If the increase in delay has been smaller for life-saving drugs, these estimates are clearly overstated.

We identify three reasons to believe that the pharmaceutical manufacturers, rather than the FDA, are more responsible for relative acceleration of important drugs. First, acceleration is observed during the 1950s, before the lengthy IND process was initiated. Second, once market importance is controlled for, scientific importance is not associated with accelerated review. Third, accelerated development appears to be a worldwide phenomenon.

This is not to say that changes in FDA policy could not significantly alter the approval rate of important new drugs. For example, recent FDA reforms allowing more widespread access to drugs treating fatal illnesses such as AIDS before the completion of clinical trials may succeed in expanding the availability of new treatments for these conditions. Likewise, the decisions by the FDA to accept data from clinical trials performed overseas and to contract with outside reviewers should further reduce development and review times.

Our results suggest that research into how firms identify important drugs and accelerate their development and approval might yield valuable insights. It is evident that firms have some ability to anticipate which drugs will be leading sellers. Given that firms can only imperfectly predict which drugs will turn out to be important, our analysis likely underestimates the true ability of firms to accelerate drug development and approval. This implies that firms already possess a tremendous ability to accelerate a given drug. Improving techniques for the early identification of the most promising compounds might therefore result in the greatest improvements in the speed with which important new drugs come to market.

There will always be concerns that accelerating the review process may compromise safety. When accelerated approval comes at the expense of thorough testing or careful review of the evidence, those concerns are likely to be particularly acute. There may even be concerns when accelerated approval is achieved only by increasing the speed of the testing and approval process. This is because these delays may provide valuable information from other countries that have approved the drug. For example, Gieringer (1985) argues that the FDA might have approved Thalidomide were it not for the disastrous

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European experience after approving the drug. Nevertheless, waiting for evidence of disaster in other countries before approving a drug for use in this country seems neither ethical nor efficient.

Of course delays in approval also have their costs, and appropriate policies must weigh the costs and benefits of accelerated testing and approval on a case-by-case basis depending on the drug's specific risks, benefits, and difficulty of testing. The results of this article suggest that some acceleration of important drugs has been taking place for years. The implications of accelerated development, testing, and review for safety are a critical area for future research.

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